COMPARADOR

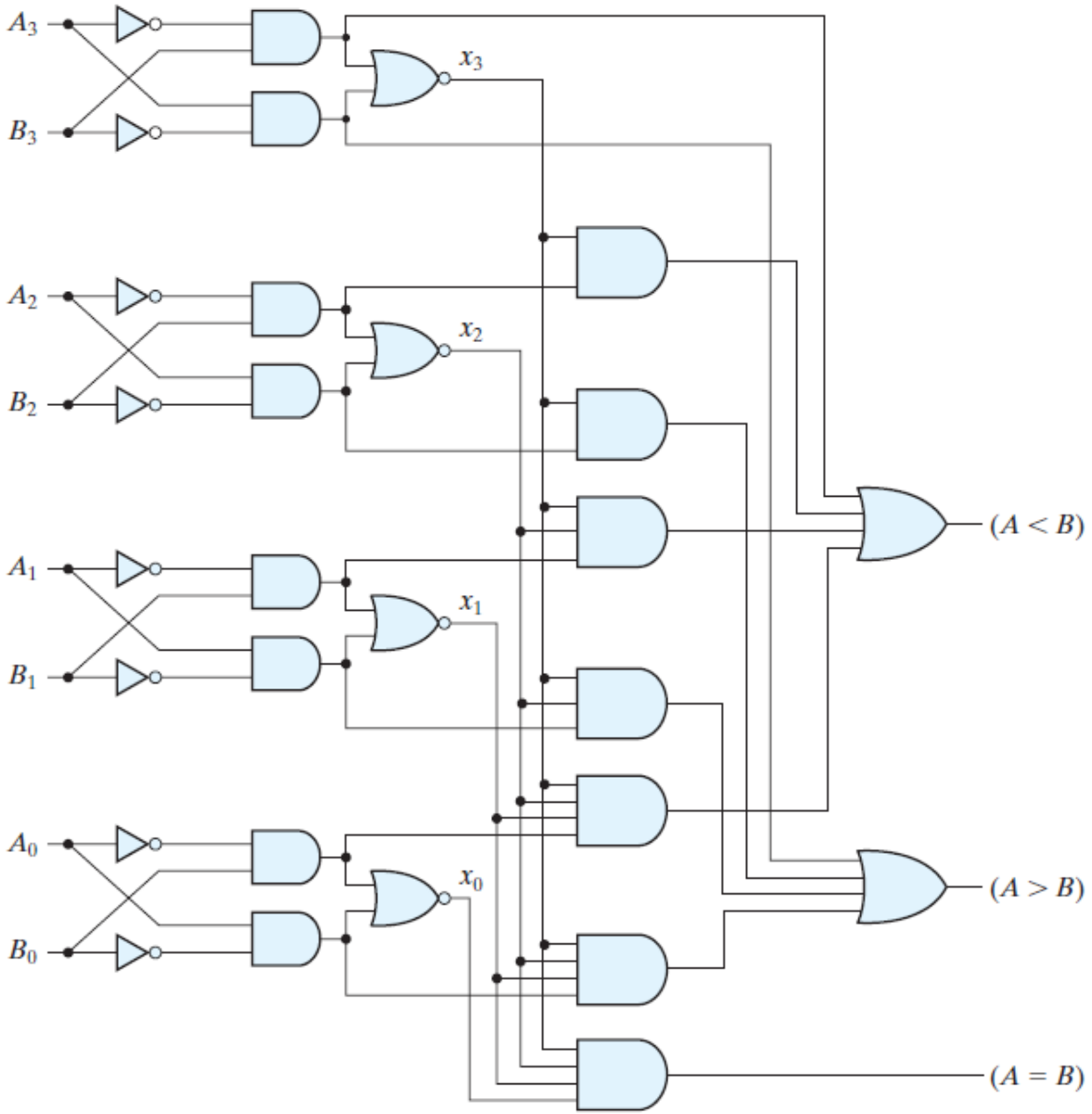
$$A = A_3\, A_2\, A_1\, A_0$$
$$B = B_3\, B_2\, B_1\, B_0$$

$$x_i = A_i B_i + A_i' B_i' \quad \text{for } i = 0, 1, 2, 3$$

$$(A = B) = x_3 x_2 x_1 x_0$$

$$(A > B) = A_3 B_3' + x_3 A_2 B_2' + x_3 x_2 A_1 B_1' + x_3 x_2 x_1 A_0 B_0'$$
$$(A < B) = A_3' B_3 + x_3 A_2' B_2 + x_3 x_2 A_1' B_1' + x_3 x_2 x_1 A' n_0 B_0'$$

COMPARADOR



DECODIFICADORES

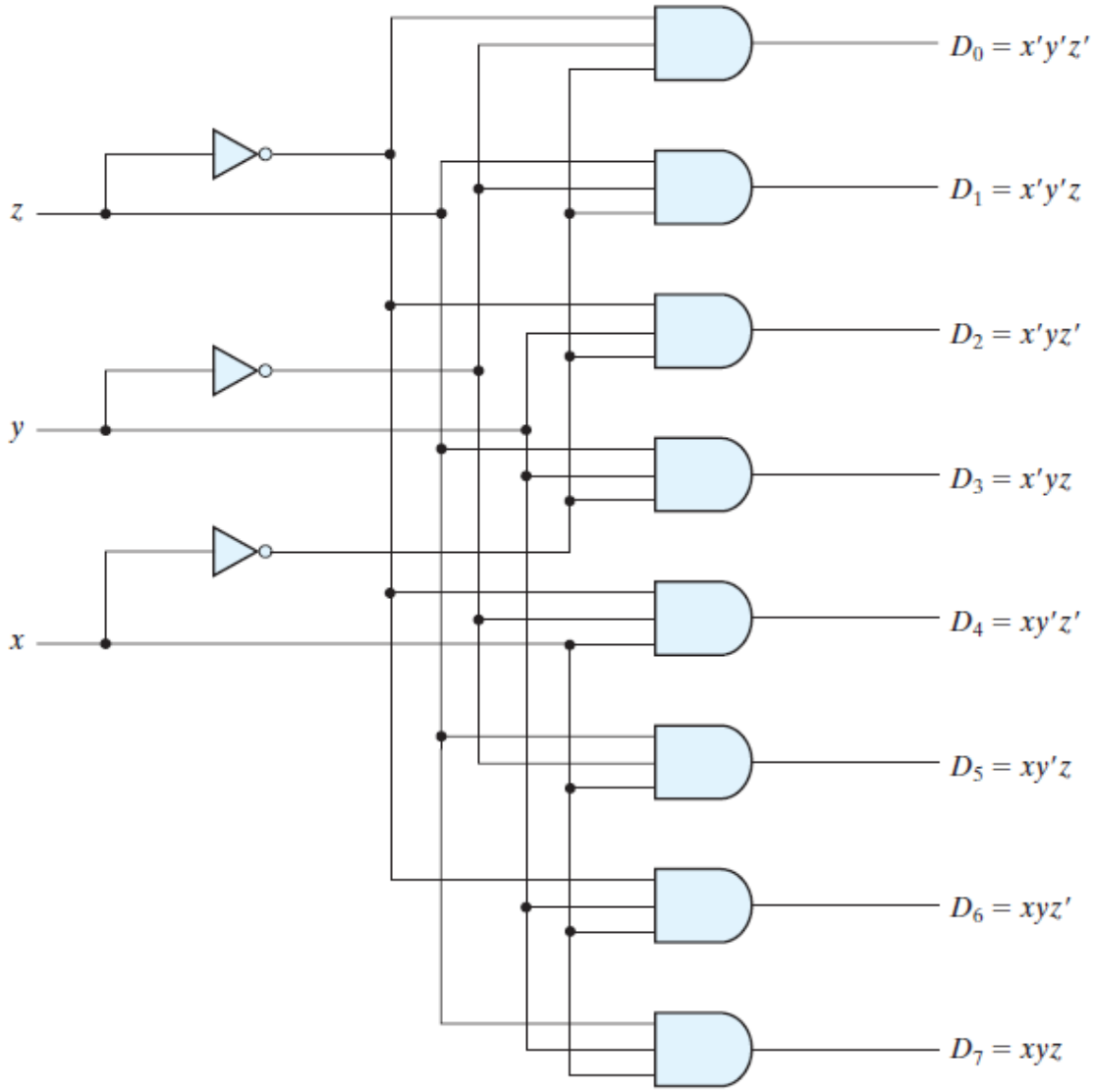
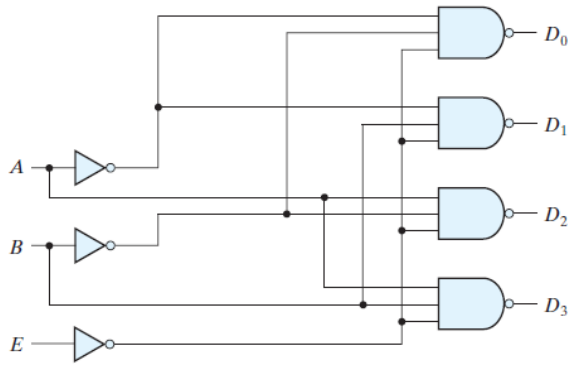


FIGURE 4.18
Three-to-eight-line decoder

DECODIFICADORES

Table 4.6
Truth Table of a Three-to-Eight-Line Decoder

Inputs			Outputs							
x	y	z	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1



(a) Logic diagram

E	A	B	D_0	D_1	D_2	D_3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	1

(b) Truth table

FIGURE 4.19
Two-to-four-line decoder with enable input

DECODIFICADORES

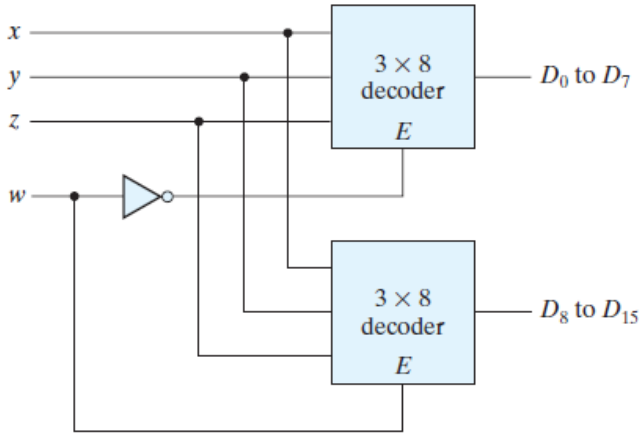


FIGURE 4.20
4 × 16 decoder constructed with two 3 × 8 decoders

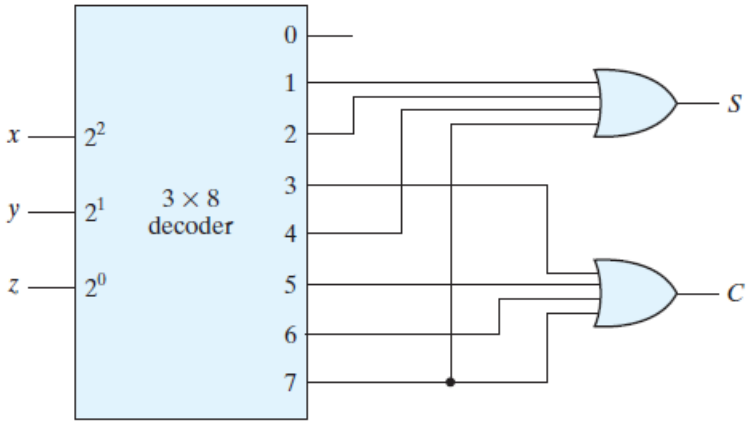


FIGURE 4.21
Implementation of a full adder with a decoder

CODIFICADORES

$$z = D_1 + D_3 + D_5 + D_7$$
$$y = D_2 + D_3 + D_6 + D_7$$
$$x = D_4 + D_5 + D_6 + D_7$$

Table 4.7
Truth Table of an Octal-to-Binary Encoder

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

CODIFICADORES

Table 4.8
Truth Table of a Priority Encoder

Inputs				Outputs		
D_0	D_1	D_2	D_3	x	y	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

$$x = D_2 + D_3$$
$$y = D_3 + D_1 D_2'$$
$$V = D_0 + D_1 + D_2 + D_3$$

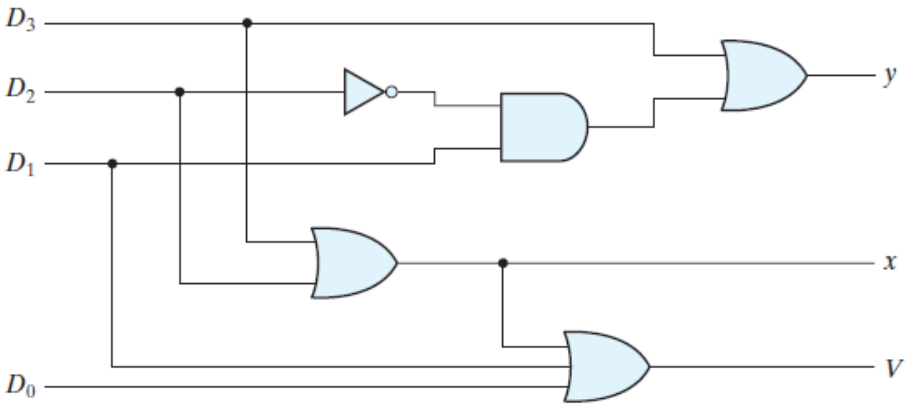


FIGURE 4.23
Four-input priority encoder

MULTIPLICADOR BINARIO

$$
\begin{array}{cccc}
 & & B_1 & B_0 \\
 & & A_1 & A_0 \\
\hline
 & & A_0B_1 & A_0B_0 \\
 & A_1B_1 & A_1B_0 & \\
\hline
C_3 & C_2 & C_1 & C_0
\end{array}
$$

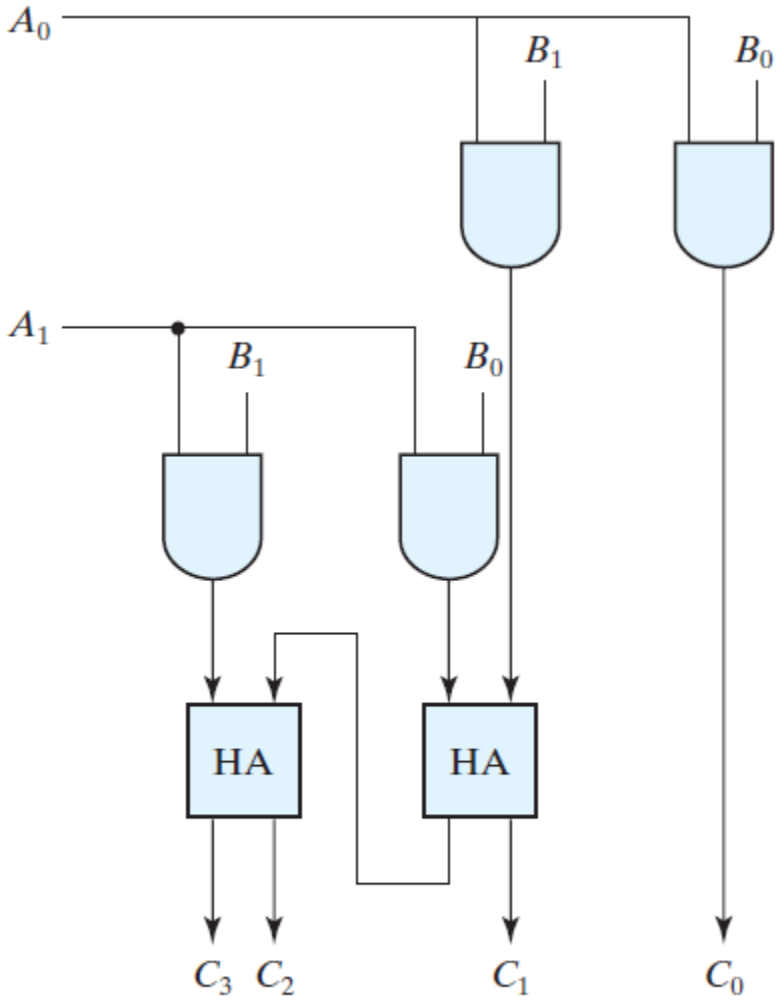


FIGURE 4.15
Two-bit by two-bit binary multiplier

MULTIPLICADOR BINARIO

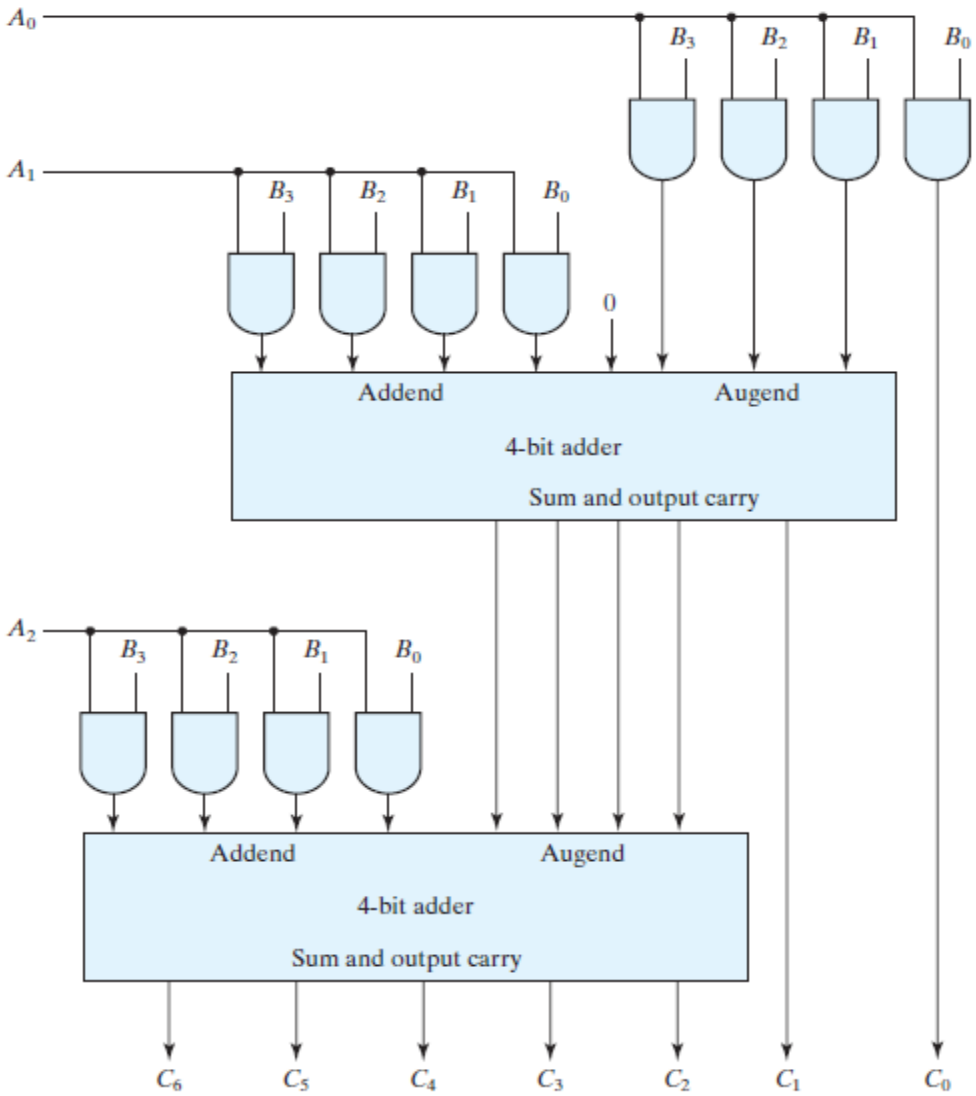


FIGURE 4.16
Four-bit by three-bit binary multiplier